

CHE 321 - Exam 1 - Sample 2

PBR_exam.m

```
1  function h = PBR_exam(t,y);
2  x = y(1);
3  p = y(2);
4  % parameters - - - - -
5  FA0 = 10
6  k = 10
7  Kc = 200
8  alpha = .07
9  epsi = 3
10 CT0 = 2
11
12 h1 = ((k/FA0)*(CT0*p*(1-x)/(1+epsi*x)))-((k/FA0)*(CT0*p*(1-x)/(1+epsi*x))*((k/FA0)*(CT0*p*(3-3*x)/(1+epsi*x)))^3
13 h2 = (-alpha/2*p)*(1+epsi*x)
14 h = [h1;h2];
```

PBR_Examsim.m

```
1  tspan = [0,10];
2  y0 = [0,1];
3  [t,y] = ode45(@PBR_exam,tspan,y0);
4  x = y(:,1);
5  p = y(:,2);
6  plot(t,x);
7
8  plot(t,y)
```